

# Secure Face Registration & Verification Framework

Calibration & Implementation Engineering Report (Days 6–9)

**Prepared For:** Face Liveness System Baseline Calibration

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**Status:** Fully Calibrated & Verified in Git repository

# 1. Day 6 — Environment Setup & Dataset Strategy

Day 6 established a clean, reproducible engineering environment and compiled the baseline development datasets required to build and validate the registration and verification pipeline.

## A. Environment Setup & Dependency Resolution

**Python Version:** Initialized a virtual environment running Python 3.11.9 (releasing version compatibility over Python 3.12+ for MediaPipe and DeepFace).

**Disk Optimization:** Free'd up 6 GB of stale downloads on E: drive. Installed libraries via pip install --no-cache-dir to protect a limited C: drive cache size.

**Deprecation Adaptations:** MediaPipe 0.10.15+ deprecated legacy solutions. Migrated sanity checking and pose mesh algorithms to use the modern MediaPipe Tasks API. Resolved Keras 3 compatibility issues under TF 2.21 by installing the tf-keras compatibility bridge.

## B. Core Dataset Specifications

To construct a balanced 5,000-image development dataset, we downloaded and sampled files via kagglehub:

| Dataset              | Kaggle Slug                          | Sample Count                                                  | License & Terms                      |
|----------------------|--------------------------------------|---------------------------------------------------------------|--------------------------------------|
| CFP Dataset          | chinafax/cfpw-dataset                | 2,000 images (500 identities, 2 frontal + 2 profile each)     | Standard research use license        |
| CelebA-Spoof (small) | trainingdatapro/celeba-spoof-dataset | 2,998 images (1,500 real augmented, 1,498 spoof video frames) | Non-commercial research only license |

## C. Self-Data Collection Statistics

**Total Captured:** 38 images inside data/self\_collected/session\_1/:

- \* **Normal Poses:** 5 Frontal, 6 Left Pose, 7 Right Pose.
- \* **Bad Quality:** 12 images (including too dark, overexposed, and blurred).
- \* **Attacks:** 8 images (including printed photo, video playback, and multiple faces).

**Staging Issues:** Initial console inputs (using Python's input()) blocked the OpenCV webcam thread and caused the window to freeze. This was resolved by implementing a custom, non-blocking on-screen state menu driven entirely by OpenCV's window key events (`1-5` keys).

## 2. Day 7 — Brightness and Blur Calibration

Day 7 focused on establishing standard mathematical thresholds to evaluate image exposure (brightness) and sharpness (blur), rejecting low-quality inputs before running costly face recognition steps.

### A. Rationale and Mathematical Approach

\* **Brightness Assessment:** Images are converted to grayscale and averaged using `cv2.mean()`. This yields a single baseline value (0–255) representing overall exposure.

\* **Blur (Sharpness) Assessment:** Variance of the Laplacian operator (`cv2.Laplacian().var()`) is computed. High-contrast edges yield high variance (sharp image), while gradual, blurred pixel transitions yield low variance (blurry image).

### B. Calibration Results and separation

The calibration script analyzed the self-collected images and produced clear separation ranges:

| Category | Count | Brightness Min/Mean/Max | Blur (Sharpness) Min/Mean/Max |
|----------|-------|-------------------------|-------------------------------|
| front    | 5     | 137.9 / 147.1 / 152.7   | 1087.7 / 1202.8 / 1494.7      |
| left     | 6     | 136.4 / 146.4 / 149.9   | 1080.8 / 1221.6 / 1338.5      |
| right    | 7     | 140.5 / 143.6 / 147.5   | 1207.6 / 1227.1 / 1258.3      |
| bad_dark | 2     | 10.6 / 10.7 / 10.8      | 10.6 / 12.6 / 14.7            |
| bad_blur | 3     | 140.2 / 149.5 / 155.9   | 530.8 / 805.6 / 965.5         |

### C. Threshold Calibration Decision

**BRIGHTNESS\_MIN = 100:** Rejects dark images (~10.7) and safely passes good poses (136.4+).

**BRIGHTNESS\_MAX = 220:** Rejects overexposed images and high glare.

**BLUR\_MIN = 1000:** Rejects motion blurred images (530.8 to 965.5) and passes all sharp poses (1080.8+).

**Result:** 100% sorting accuracy achieved. Logged to `data/day7_quality_results.csv`.

### 3. Day 8 — Pose, Position, and Single-Face Validation

Day 8 implemented geometry-based validations: verifying that exactly one person is present, ensuring they are centered and close enough, and measuring head pose angles using 3D-to-2D perspective projection.

#### A. Mathematical Modeling & PnP Ambiguity Resolution

We utilized OpenCV's solvePnP algorithm mapping four MediaPipe landmarks: Nose Tip (1), Chin (152), Right Eye Outer (33), and Left Eye Outer (263) against a generic 3D model. Two critical bugs were resolved:

- 1. **Coordinate Inversion:** MediaPipe's Y-axis increases downwards while standard 3D models increase upwards. This originally caused roll outputs to read near 180°. We aligned the 3D coordinate signs (Y-down, X-right).
- 2. **Dual Solution Ambiguity:** With only 4 points, the iterative solver can converge to a flipped upside-down solution. We solved this by passing a forward-facing initial guess (rvec = 0, tvec = focal\_length) and setting useExtrinsicGuess=True. Roll angles successfully stabilized near 0°.

#### B. Calibration Results and Threshold Alignments

The calibration script processed the 38 self-collected images and yielded the following ranges:

| Category | Face Area Ratio Range | Yaw Angle Range  | Pitch/Roll Max             |
|----------|-----------------------|------------------|----------------------------|
| front    | 0.039 to 0.126        | -23.2° to 18.8°  | Pitch: 13.1°   Roll: 2.6°  |
| left     | 0.053 to 0.095        | -32.0° to -58.9° | Pitch: 14.9°   Roll: 8.3°  |
| right    | 0.051 to 0.061        | 37.5° to 63.4°   | Pitch: 42.2°   Roll: 43.2° |

#### C. Calibrated Threshold Decision

**YAW\_FRONTAL\_MAX = 25.0°:** Natural sitting postures measured up to 23.2° yaw. Widened from 15.0° to prevent false rejections.

**YAW\_PROFILE\_MIN = 25.0° / MAX = 65.0°:** Widened the target window from 15–35° to 25–65° to capture the user's actual, deeper profile turn angles.

**PITCH\_MAX = 35.0°:** Widened from 20.0° to accept tilting.

**MIN\_FACE\_AREA\_RATIO = 0.03:** Lowered from 0.08 because a natural sitting distance (50-70cm) covers 4% to 12% of the frame.

## 4. Day 9 — Occlusion Detection Approximation

Day 9 implemented an occlusion check to detect when key face regions (eyes, nose, mouth) are covered by hands, hair, or masks, which would compromise face recognition matching.

### A. Rationale and Structural Limitations

MediaPipe Face Mesh does not expose a per-landmark visibility confidence score. Therefore, we implemented a **two-signal approximation** as documented in the Approach & Design Document:

1. **Face Detector Confidence Score:** Obscuring the face causes the detector's confidence score to drop. If confidence falls below `DETECTION_CONFIDENCE_MIN` = 0.80, occlusion is flagged.
2. **Local Texture Variance:** Extracts a 24x24 pixel patch around the left eye, right eye, nose tip, and mouth region, and computes local Laplacian variance. Flat regions (like hands or masks covering skin) yield extremely low variance.

### B. Calibration Observations & Mixed Separation

Our calibration run revealed mixed but useful separation characteristics: \* `bad_occlusion_001.jpg`: Successfully failed the occlusion check. The mouth region was covered, and its local variance dropped to `14.7` (failing our `25.0` threshold). \* `bad_occlusion_002.jpg`: Passed the check (no occlusion flagged). The hand covering the mouth had visible lines and skin texture, yielding a variance of `84.3` (which is indistinguishable from smooth visible skin on good frontal poses like `front_005` which measured `77.1`). \* **Zero False Positives:** All good poses (frontal/left/right) successfully passed the check (min variance observed was 77.1), confirming the occlusion threshold avoids false rejections.

### C. Calibration Threshold Decision

**Local Patch Variance Limit = 25.0:** Safely flags flat covered regions without triggering false positives on visible skin.

**DETECTION\_CONFIDENCE\_MIN = 0.80:** Rejects low-confidence detections.

**Conclusion:** Documented as an approximation. The system successfully flags solid, flat cover occlusions, but cannot reliably classify textured covers (such as detailed skin/lines on hands). Pushed to `data/day8_9_quality_results.csv`.